AI-Powered Nail Art Preview - Technical Implementation

Objective

Allow users to upload a photo of their nails and preview selected nail art designs applied realistically using AI techniques. The system should detect nail areas, apply design overlays with image-to-image transformation, and return a photorealistic result.

Step 1: Nail Detection and Segmentation

- Use Mediapipe or Meta's Segment Anything Model (SAM) to identify and extract nail areas from the uploaded image.
- Output: A segmentation mask representing nail regions.

Step 2: Design Application (Image-to-Image AI)

- Apply selected nail art design using ControlNet and Stable Diffusion.
- Use segmentation mask as guidance input (ControlNet) to ensure accurate placement.
- Generate new image with the design realistically blended onto nail area.

Step 3: Backend API Architecture

- Use Flask or FastAPI to build an API that handles image upload and processing.
- The API receives user photo and selected design, processes it with the AI model, and returns generated image URL.
- Optionally use Replicate or Render for running models in the cloud.

Technologies and Tools

- Segment Anything Model (SAM) - Nail area segmentation
- ControlNet + Stable Diffusion - Design transformation
- Python + Flask/FastAPI - Backend API
- Firebase Storage or Cloudinary - Hosting user-generated and result images
- Mobile App: React Native or Flutter frontend integration

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Optional Enhancements

- Skin tone detection for color harmony
- Suggest similar styles using AI
- Save and share preview image
- Personalize results with user preferences

Conclusion

This architecture allows a mobile application to leverage AI in providing a hyper-personalized, realistic nail art preview experience. Scalable, extensible, and suitable for MVP-to-product expansion.